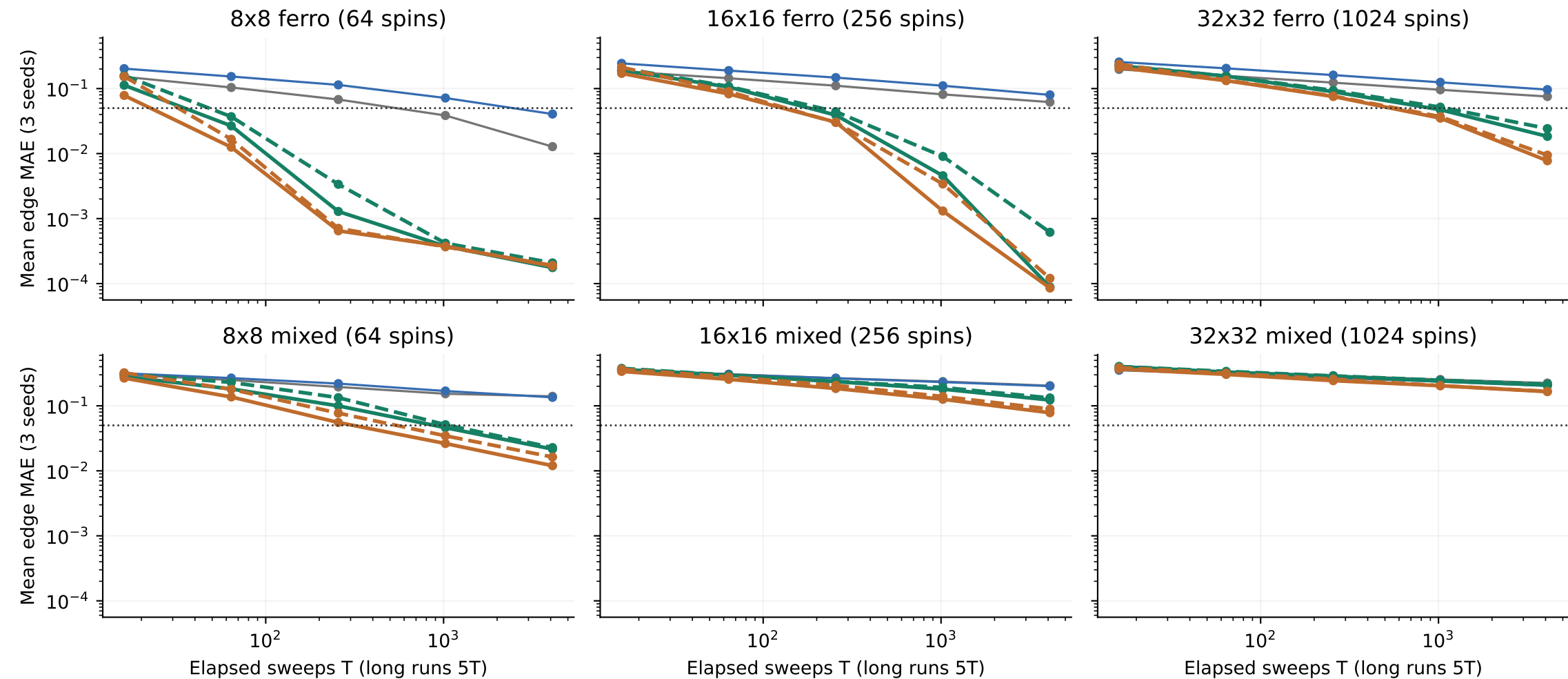


Past 64 spins only tempering reaches the threshold; mixed grids stay out of reach

- Long chain
- Five cold chains
- Five replicas, $k=1$
- Five replicas, $k=4$
- Nine replicas, $k=1$
- Nine replicas, $k=4$



Dotted line: qualification threshold 0.05, which must hold at every larger budget. Exact Kac-Ward references.
CPU software simulation, two-colour block Gibbs at beta 4, 16 trials per cell; no hardware claim.